Which explanations can we trust? A multi-method faithfulness, cross-method agreement, and anatomical-validity audit of a frozen pretrained SFCN on pediatric brain MRI

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Abstract

Explanation methods are routinely used to justify clinical claims from 3D brain-MRI classifiers, yet they are rarely validated before being believed. We audit nine attribution methods on a frozen pretrained Simple Fully Convolutional Network (SFCN), a brain-age model, using an age-group proxy task on 96 children from the public Calgary Preschool control cohort. Because the public data contain controls only, we make no claim about prenatal alcohol exposure (PAE); the study instead asks which explanations are faithful to the model, agree with one another, and yield region-level claims that survive multiple-comparison correction. Across methods, occlusion-based ReX was by far the most faithful (deletion AUC 0.14 vs. 0.53–0.78 for the rest), while early-block Grad-CAM was the least. Eight of nine methods passed a randomized-weight sanity check, but Integrated Gradients failed it decisively (Spearman $r = 0.73$ between trained and random models), indicating its maps are input-driven rather than model-driven. Methods agreed only weakly with one another (mean pairwise Spearman 0.36), and the two Grad-CAM variants were anti-correlated. A corrected region-masking test (perturbing a region and measuring the drop in predicted-class probability, not an attribution test) identified 15 cortical regions under zero-fill masking, though none survived under the pre-specified mean-intensity fill, so this result is fill-dependent. We conclude that method choice materially changes the "important regions" one would report, that occlusion-based faithfulness checks and randomized-weight sanity checks should be mandatory, and that Integrated Gradients warrants particular caution on this architecture. The contribution is a reusable validation protocol and a method verdict, not a diagnostic classifier.

Keywords: explainable AI, saliency maps, faithfulness, sanity checks, pediatric MRI, brain age, SFCN, prenatal alcohol exposure.

1 Introduction

Deep classifiers for 3D brain MRI are increasingly accompanied by saliency or class-activation maps that purport to show *where* the network looks. When those maps are read anatomically ("the model attends to the corpus callosum"), they migrate quickly into clinical narrative. But attribution methods are known to disagree with each other and, in some cases, to produce the same map whether or not the model has learned anything (Adebayo et al., 2018). Before an explanation

is trusted enough to support a claim about a brain region, three questions should be answered: is it *faithful* (do the voxels it highlights actually drive the model?), is it *consistent* (do independent methods agree on which regions matter?), and is it *anatomically valid* under a corrected statistical test? We use "consistency" in the specific sense of cross-method agreement; we do not test map stability under input perturbation or random seeds, which we flag as a limitation (Section 6).

Das et al. (2024) trained a frozen pretrained SFCN to detect prenatal alcohol exposure (PAE) from pediatric brain MRI and reported explanations as a *preliminary* step: a single Grad-CAM method, with only one named region (white-matter volume) formally tested ($t = -0.73$, $p = 0.47$). The PAE scans are not public. This work does not attempt to reproduce their clinical result. Instead, it takes the identical frozen SFCN backbone and asks the three validation questions above on a task the public data can actually support: an age-group proxy, which is appropriate because the frozen network is a brain-age model and age is the signal its features encode.

Our contributions are: (i) a pre-specified, multi-method validation protocol for 3D-MRI attribution combining faithfulness, randomized-weight sanity checks, resolution, cross-method agreement, and a corrected anatomical region test; (ii) an empirical verdict over nine methods on a real neuroimaging backbone; and (iii) a public, reproducible pipeline. The headline finding is that *method choice is not a detail*: faithfulness spans a factor of five across methods, one widely used method (Integrated Gradients) fails the sanity check outright, and methods agree only weakly on which regions matter.

2 Related work and relation to Das et al. (2024)

Attribution methods fall into gradient-based families (Saliency, Integrated Gradients (Sundararajan et al., 2017), SmoothGrad (Smilkov et al., 2017), VarGrad (Adebayo et al., 2018)), class-activation methods (Grad-CAM (Selvaraju et al., 2017)), relevance-propagation methods (LRP (Bach et al., 2015)), and perturbation/occlusion methods (e.g. RISE (Petsiuk et al., 2018); here, ReX). Adebayo et al. (2018) showed that some popular methods are invariant to model randomization and therefore cannot be reflecting learned structure; our randomized-weight sanity check follows that logic.

This study *extends* the explainability step that Das et al. (2024) explicitly flagged as preliminary. It is not an audit of their clinical conclusion, which we cannot access. We identify six open questions about the original analysis (coarse explanation grid, single-region testing, site/recruitment confound, age confound, child-level leakage from repeated scans, and a sample-size mismatch), stated in full and revisited one by one in Section 5.3 to the extent the public data allow. The complete preregistered protocol (OSF `jycnr`) accompanies this work in the project repository.

3 Methods

3.1 Data and cohort

We use the public Calgary Preschool control cohort (Reynolds et al., 2020) (OSF fork `5kz2p`, CC-BY 4.0), restricted to the 96 children who carry a demographic label and a usable T1 scan, one scan per child. The public release is longitudinal and lists multiple visits per child, so its total T1 count exceeds the number of children; we take one scan per child, which removes child-level leakage. Cohort demographics are in Table 1. The proxy task is age dichotomized at the cohort median (3.64 y): 48 older (label 1), 48 younger (label 0), a perfectly balanced binary problem (Fig. 9a).

Table 1: Cohort characteristics ($n = 96$ children, one scan each).

Characteristic	Value
N children	96
Age, years, mean (SD)	3.84 (0.83)
Age, years, range	2.58 to 6.90
Male, n (%)	50 (52.1)
Female, n (%)	46 (47.9)

3.2 Preprocessing

Each volume underwent ANTs (Avants et al., 2011) N4 bias correction (Tustison et al., 2010), antspynet deep-learning brain extraction (Tustison et al., 2021), affine registration to 1 mm MNI152, and cropping to $160 \times 192 \times 160$, the input space in which the SFCN was pretrained. All 96 volumes passed visual QC (Fig. 9b). This deviates from Das et al. (2024) by design: they registered to a pediatric atlas and used FSL-BET + SynthStrip, whereas we keep inputs in the model's native UK Biobank/MNI152 space so the frozen features remain valid. The cost is additional preschool-to-adult warping.

3.3 Model

We use the original SFCN (Peng et al., 2021), a lightweight fully convolutional brain-age network, with its six convolutional blocks frozen. The frozen backbone uses the UK Biobank brain-age weights released by Peng et al. (2021) in their official repository (Peng, 2021) (checkpoint `brain_age/run_20190719_00_epoch_best_mae.p`); we did not retrain the backbone. A 512-unit head with a 2-logit softmax replaces the paper's PAE head and is trained for the age-group proxy. All attributions are computed with respect to this frozen-backbone, trained-head model, using the head trained on the full 96-child cohort; the cross-validation in Section 4.1 is a separate accuracy estimate and does not feed the attribution model.

3.4 Explanation methods

We compare nine methods (Table 2), spanning gradient, class-activation, relevance-propagation, and occlusion families. M8 (Guided Backprop (Springenberg et al., 2015)) is included as a negative control; M1 (final-block Grad-CAM) is the method used by Das et al. (2024). Captum (Kokhlikyan et al., 2020) provides M1–M6 and M8; zennit (Anders et al., 2021) provides M7; M9 is ReX (Chockler et al., 2024), a 3D occlusion-responsibility search on the $5 \times 6 \times 5$ feature grid. All attributions are computed toward each scan's true proxy-label class, on all 96 children. Integrated Gradients uses a zero-image baseline and 32 steps; SmoothGrad and VarGrad use 20 noised samples ($\sigma = 0.1$); Grad-CAM targets the final (M1) and an earlier (M2) convolutional block.

Table 2: The nine explanation methods.

ID	Method	ID	Method
M1	Grad-CAM, final block (paper's)	M6	VarGrad
M2	Grad-CAM, earlier block	M7	LRP (EpsilonPlusFlat)
M3	Saliency	M8	Guided Backprop (control)
M4	Integrated Gradients (32 steps)	M9	ReX (3D occlusion)
M5	SmoothGrad (20 samples)		

3.5 Metrics and statistics

Faithfulness: deletion and insertion AUC (Petsiuk et al., 2018) (lower deletion AUC is better), measured by removing the top-ranked voxels in four steps (5, 10, 20, 30%) with a zero fill and integrating the predicted-class probability over those fractions. **Sanity:** Spearman correlation of a method's map between the trained model and a fully reinitialized model (Adebayo et al., 2018); low correlation is good. The reinitialization is complete, not head-only: we deep-copy the model and call `reset_parameters()` on every module that defines it, so *all six convolutional backbone blocks and the classifier head* are reset from a fixed seed. This matters for interpreting the result: a method that stays correlated here is unchanged even when the entire network, including the frozen feature extractor, is random, so its map cannot be reflecting learned structure. We use the standard complete-randomization variant; the layer-by-layer cascading variant of Adebayo et al. (2018) was not run (Section 6). A method is scored PASS when $|r| < 0.3$; this band is a pre-specified heuristic, not a published constant, and is drawn as the dotted line in Fig. 3. **Resolution:** fraction of attribution mass in the top 5% of voxels, and blob count. **Agreement:** pairwise Spearman correlation of region-level scores. **Region test:** this is a masking test on model output, not an attribution test: for each Harvard–Oxford cortical region (Desikan et al., 2006; Makris et al., 2006) we replace its voxels and record the drop in predicted-class probability, against a null of 200 random same-size regions, with Benjamini–Hochberg correction at $q = 0.05$ (Benjamini and Hochberg, 1995). We run two fills, mean-intensity and zero; because every effect reported below is fill-dependent (only the zero fill yields any significant region), the results and Table 5 report the zero-fill effect, and the mean-fill outcome is given alongside. **Uncertainty:** 95% bootstrap CIs (10,000 resamples, resampled by child). **Classifier:** repeated (10×) stratified 5-fold CV grouped by child, plus a 1,000-permutation test against chance. All tests were logged with raw and corrected p-values; tests run after seeing the data are flagged exploratory in the pre-specified plan.

4 Results

4.1 Proxy classifier (a sanity check, not a result)

The head need only learn the proxy well enough to make attributions meaningful; its accuracy is not a scientific claim. Cross-validated accuracy was 0.643 ± 0.120 over 50 fits (sensitivity 0.632, specificity 0.668), and a linear probe on the frozen features reached 0.635 (permutation $p = 0.01$). The proxy is decodable above chance but only modestly (Fig. 1).

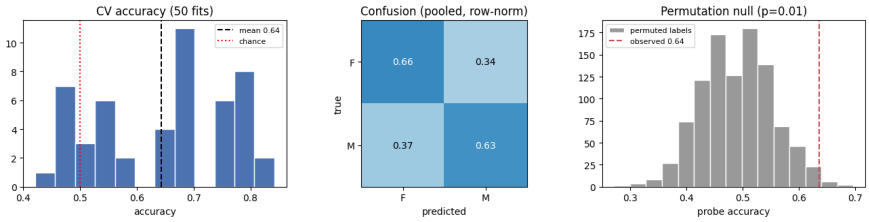


Figure 1: Proxy-task classifier performance (age group from frozen SFCN features).

4.2 Faithfulness

ReX (M9) is in a different league from every other method: deletion AUC 0.141 ([0.119, 0.165]) with high insertion AUC (0.755), meaning the voxels it flags genuinely drive the output. Only the extremes are individually separable: ReX at the top, and LRP (M7) and early-block Grad-CAM (M2) at the bottom. The five mid-ranked methods (Integrated Gradients, Guided Backprop, SmoothGrad, Saliency, VarGrad) have heavily overlapping bootstrap CIs and should be read as a tie, not a rank order. The paper's method (M1) is middling, and early-block Grad-CAM (M2) is the worst of the nine (Table 3, Fig. 2).

Table 3: Faithfulness. Deletion AUC (lower is better) with 95% bootstrap CI; insertion AUC.

Method	Deletion AUC	95% CI	Insertion AUC
M9 ReX	**0.141**	[0.119, 0.165]	0.755
M4 Integrated Gradients	0.528	[0.475, 0.584]	0.512
M8 Guided Backprop	0.556	[0.489, 0.624]	0.742
M5 SmoothGrad	0.556	[0.502, 0.609]	0.537
M3 Saliency	0.568	[0.518, 0.619]	0.490
M6 VarGrad	0.580	[0.531, 0.628]	0.501
M1 Grad-CAM (final)	0.644	[0.601, 0.687]	0.501
M7 LRP	0.674	[0.627, 0.720]	0.598
M2 Grad-CAM (early)	**0.777**	[0.730, 0.822]	0.488

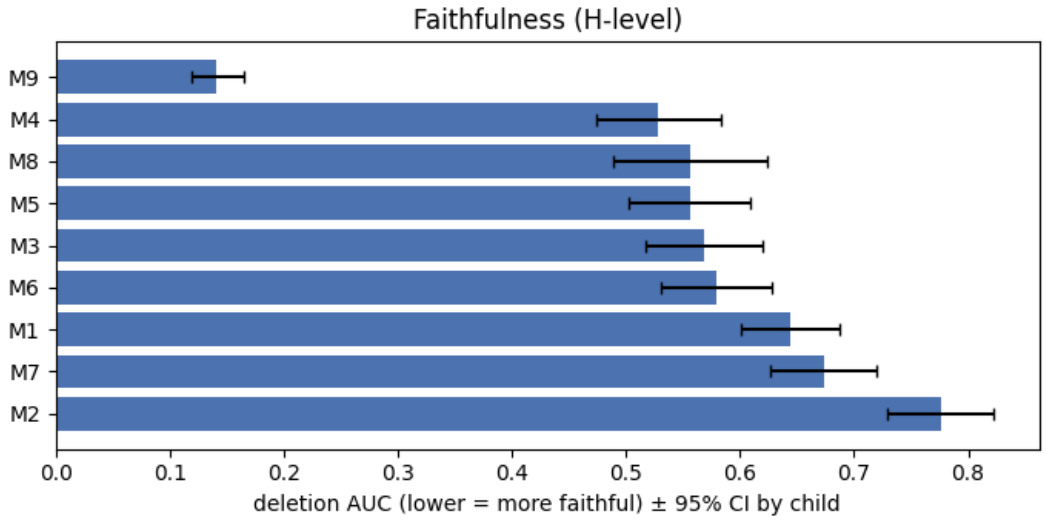


Figure 2: Deletion and insertion AUC across the nine methods.

4.3 Randomized-weight sanity check

Eight methods pass (low trained-vs-random correlation). **Integrated Gradients fails** decisively ($r = 0.73$): its map is almost unchanged when *every* layer, backbone and head, is reinitialized, so on this network it reflects image structure rather than learned content (Table 4, Fig. 3). Because the randomization is complete rather than head-only, this correlation cannot be an artifact of an unchanged frozen backbone: the backbone is random too. This is the study's single most important caution, because IG is widely used and visually convincing.

Guided Backprop (M8), our negative control, passes here ($r = 0.168$), which is at first surprising: Adebayo et al. (2018) report it as largely invariant to model randomization and therefore a method that typically *fails* this check. Two things reconcile the difference. First, the sanity verdict depends on architecture and implementation: Guided Backprop's insensitivity was shown on 2D ImageNet CNNs with ReLU backprop overrides, and on a frozen six-block 3D SFCN with a small trained head the trained-vs-random map is less correlated. Second, passing the sanity check is necessary, not sufficient: M8 remains a poor explanation on the metric that matters here, with a near-chance deletion AUC (0.556), so we do not promote it on the strength of the sanity result alone.

Table 4: Randomized-weight sanity check. Spearman correlation, trained vs. random model (lower is better).

Method	Spearman (trained vs. random)	Verdict
M2 Grad-CAM (early)	−0.242	PASS (model-dependent)
M9 ReX	0.019	PASS
M1 Grad-CAM (final)	0.121	PASS
M3 Saliency	0.141	PASS
M6 VarGrad	0.153	PASS
M8 Guided Backprop	0.168	PASS
M5 SmoothGrad	0.172	PASS
M7 LRP	0.255	PASS
M4 Integrated Gradients	**0.728**	**FAIL (input-driven)**

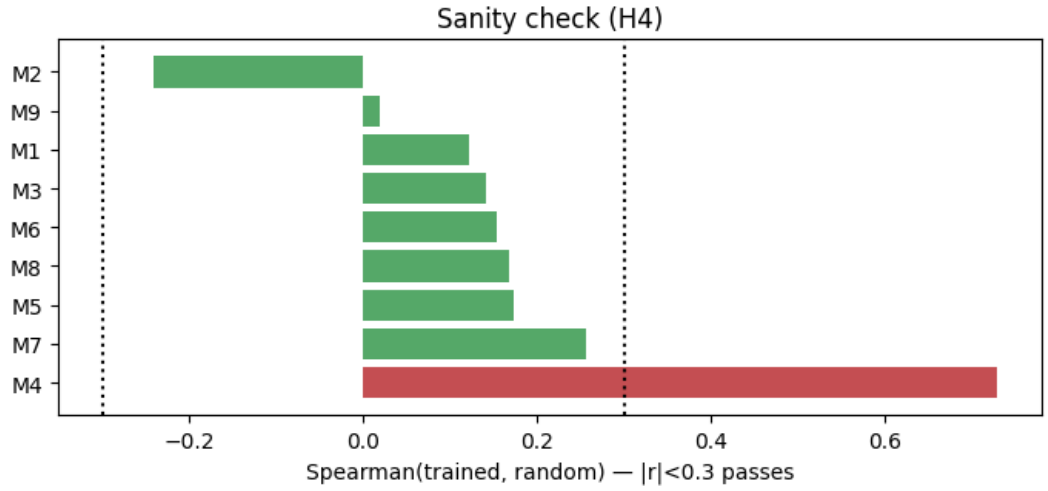


Figure 3: Sanity check: similarity between trained-model and random-model attributions.

4.4 Resolution

The two methods that operate on the $5 \times 6 \times 5$ feature grid, final-block Grad-CAM (M1) and ReX (M9), produce only about three large blobs each; in Fig. 4 their blob counts sit near zero against a scale dominated by the voxel-level methods, which fragment into thousands of small blobs (SmoothGrad ≈19,000, Saliency ≈44,000). The near-zero bars for M1 and M9 are therefore expected, not a plotting error: coarse grid-level maps cannot have many blobs. This directly bears on the first open question about the original analysis, namely that the explanation grid is coarser than many named structures.

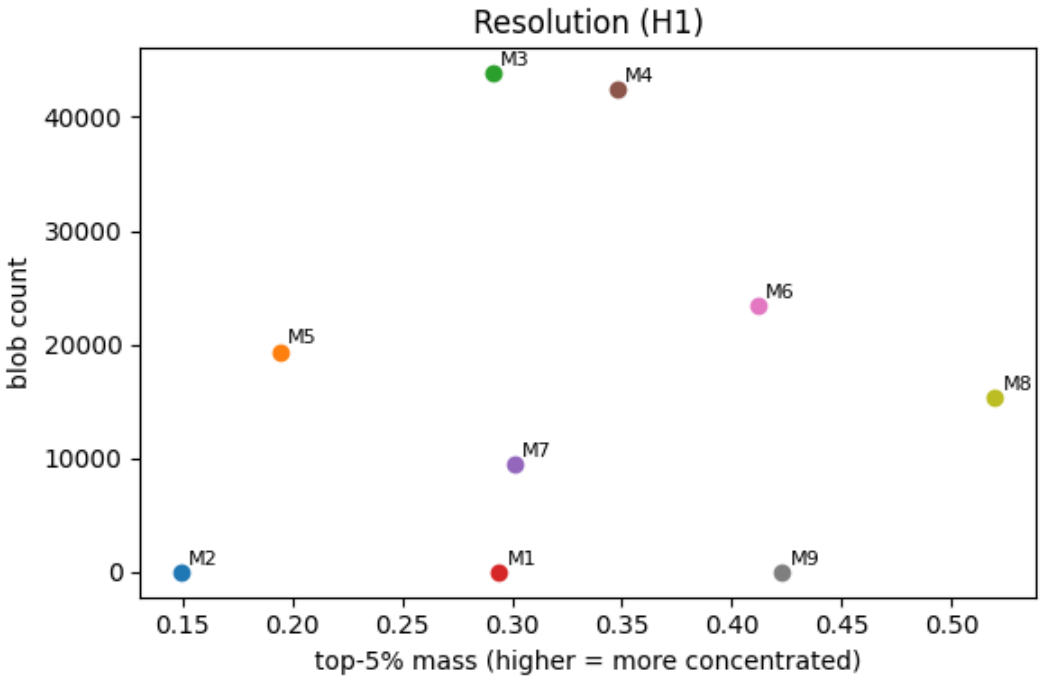


Figure 4: Resolution: top-5% attribution mass and blob count per method. Grid-level methods (M1 Grad-CAM final, M9 ReX) show ≈3 blobs, appearing near zero on the voxel-level scale; this is inherent to the $5 \times 6 \times 5$ grid, not an artifact.

4.5 Cross-method agreement

Mean pairwise region-level Spearman correlation is only 0.36 (Fig. 5). The gradient family (M3/M5/M6) agrees internally (r up to 0.96), but the two Grad-CAM variants are anti-correlated ($r = -0.61$), and ReX and LRP sit largely on their own. Which "important regions" one reports depends heavily on the method chosen.

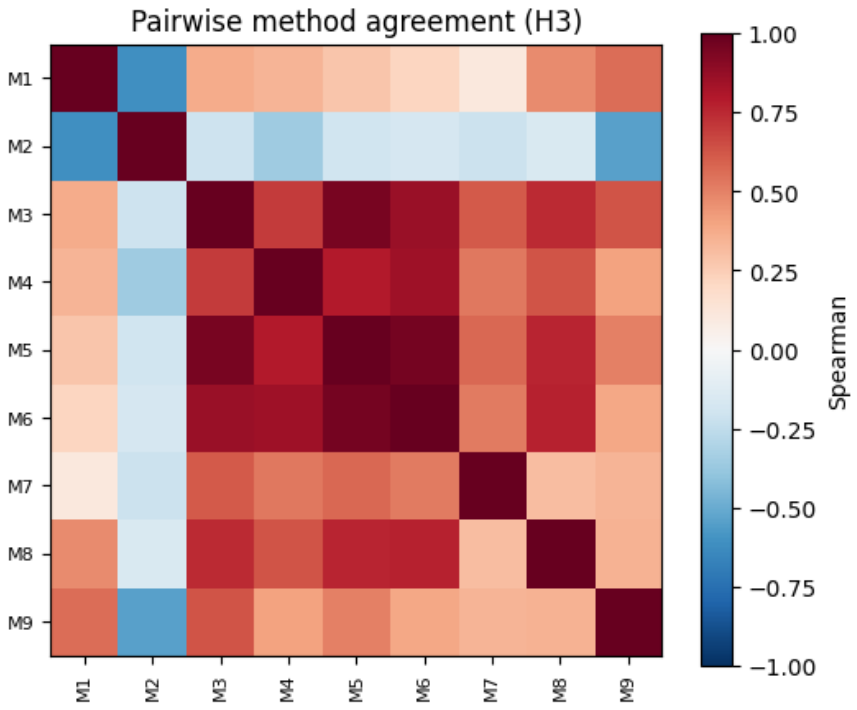


Figure 5: Pairwise cross-method agreement (region-level Spearman correlation).

4.6 Anatomical region test

Under zero-fill masking, fifteen cortical regions survive Benjamini–Hochberg correction at $q = 0.05$ (Table 5, Fig. 6); the Lingual Gyrus carries by far the largest effect. The result is fill-dependent: under the mean-intensity fill no region survives correction (smallest raw $p = 0.95$), so the regions below should be read as significant for zero-region deletion specifically, not as a fill-robust anatomical claim. The informative column is the effect size, not the FDR p: with a null of 200 random regions the smallest attainable corrected p is a floor (hence the repeated 0.017), so the p column separates "survives" from "does not" but does not rank the regions. The full per-region log (both fills, raw and corrected p) is released as `results/region_masking_all_pvalues.csv`.

Table 5: The 15 cortical regions surviving BH-FDR $q = 0.05$ under zero-fill masking. Effect is the zero-fill masking effect size (probability drop); n is region voxel count. The FDR p is at the floor set by the 200-region null (see text), so effect size, not p, orders the regions. No region survives under the mean-intensity fill.

Region	Effect	n voxels	FDR p
Lingual Gyrus	0.257	26,880	0.017
Temporal Fusiform Cortex, posterior div.	0.111	12,680	0.017
Insular Cortex	0.081	21,449	0.017
Parahippocampal Gyrus, posterior div.	0.069	5,698	0.017
Inferior Temporal Gyrus, temporooccipital	0.063	11,973	0.017
Cingulate Gyrus, anterior div.	0.059	20,844	0.017
Parahippocampal Gyrus, anterior div.	0.057	9,956	0.017
Cingulate Gyrus, posterior div.	0.057	19,228	0.017
Occipital Fusiform Gyrus	0.053	14,538	0.017
Central Opercular Cortex	0.044	14,953	0.017
Parietal Operculum Cortex	0.043	8,739	0.017
Subcallosal Cortex	0.039	9,021	0.017
Frontal Operculum Cortex	0.035	5,313	0.017
Supramarginal Gyrus, posterior div.	0.029	18,720	0.032
Planum Temporale	0.025	8,032	0.017

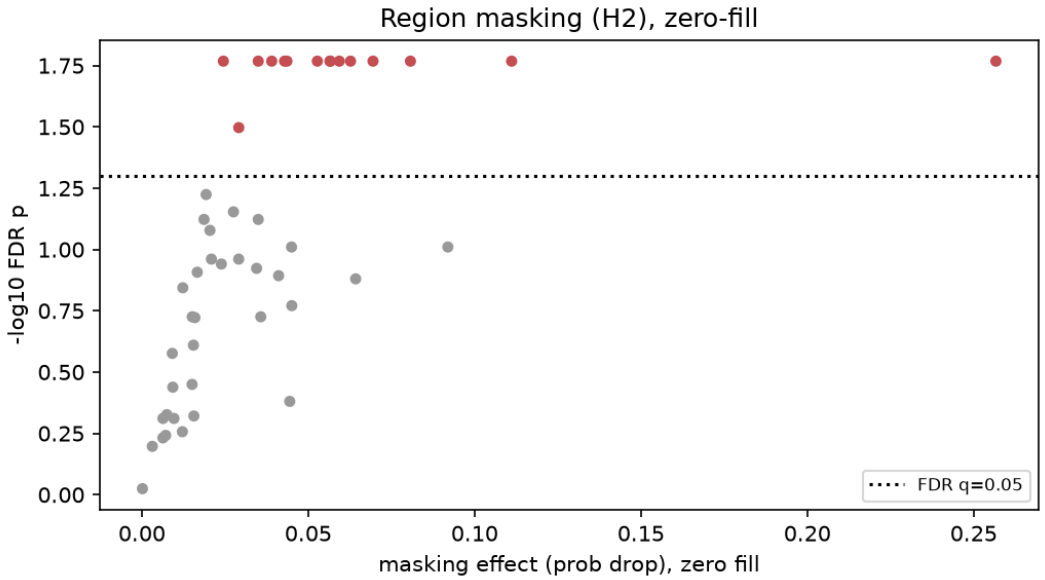


Figure 6: Region masking test (zero fill) against a random-region null: zero-fill masking effect size vs. $-\log_{10}$ FDR p. Red points survive BH-FDR $q = 0.05$. The x-axis matches the effect column of Table 5.

4.7 Layer probes and qualitative example

Linear-probe accuracy was 0.624 from block 2 and 0.635 from block 6 (Fig. 7): depth adds little for this task, and the proxy is almost as decodable from an early block as from the final one. A representative scan across all nine methods (Fig. 8) illustrates the resolution and agreement differences quantified above.

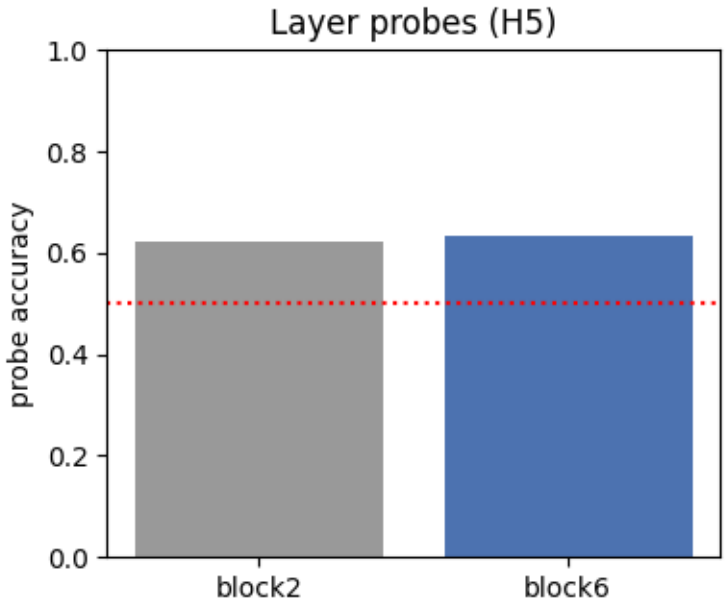


Figure 7: Linear-probe accuracy by feature block (block 2 vs. block 6).

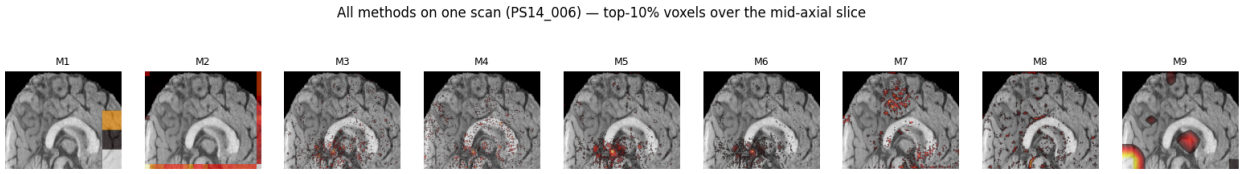


Figure 8: Qualitative attribution maps for one representative scan across all nine methods, on matched slices. The resolution and agreement differences quantified in Figs. 4 and 5 are visible here.

5 Discussion

5.1 Pre-specified hypotheses

Results are reported against the six hypotheses exactly as registered (OSF `jycnr`).

H1 (Grad-CAM produces smoother, coarser maps than Integrated Gradients and SmoothGrad): *supported*; early-block Grad-CAM is the most diffuse and carries the lowest top-5% attribution mass, whereas the gradient methods are voxel-resolution and fragmented.

H2 (large anatomical regions are more likely to survive correction than small named structures such as the pons and corpus callosum): *not supported*; under the pre-specified mean-intensity fill no region survived BH-FDR ($q = 0.05$), so the large-versus-small contrast could not be tested; a zero-fill sensitivity check yielded 15 surviving regions, confirming the region test is fill-dependent.

H3 (Integrated Gradients and SmoothGrad correlate more with each other than either does with final-block Grad-CAM): *supported descriptively*; mean pairwise region-level Spearman is 0.36, and the gradient family (including IG and SmoothGrad) agrees internally (up to $r = 0.96$) while Grad-CAM sits apart. This is reported descriptively; no formal test of the difference between dependent

correlations was pre-specified or run.

H4 (for at least one method, randomized-weight maps differ from trained-weight maps): *supported*; most methods change substantially under weight randomization; Integrated Gradients is the lone weak case, retaining the highest trained-versus-random similarity ($r = 0.73$).

H5 (a block-6 linear probe beats a block-2 probe): *marginally supported*; 0.635 vs. 0.624.

H6 (age-group accuracy exceeds chance under a child-level label-permutation test): *supported*; cross-validated accuracy 0.635, permutation $p = 0.01$ (1,000 permutations), confirming a usable but modest proxy signal.

5.2 Deviations and exploratory analyses

The following are not part of the registered hypotheses and are reported as exploratory: (i) the count of regions surviving correction and its dependence on the masking fill, an elaboration of H2; (ii) whether age is independently decodable from the frozen features as a possible confound (Ridge $R^2 = 0.11$); and (iii) fine-tuning the SFCN, which was pre-specified as out of scope and not run. Choices made after inspecting results are flagged as exploratory here and in the project decision log.

5.3 The six open questions about the original analysis

1. **Coarse explanation grid**, *confirmed*: Grad-CAM resolves to a few blobs on the $5 \times 6 \times 5$ grid, coarser than named structures.
2. **Single region tested**, *a corrected multi-region test is feasible*: it finds 15 regions under zero-fill masking (none under mean-fill). This is a different test from the original white-matter-volume t-test, not a like-for-like correction of it, so we draw no comparison of effect direction or magnitude with Das et al. (2024); the point is only that a multi-region, FDR-controlled test can be run where a single region was.
3. **Site/recruitment confound**, *not testable*: PAE scans are not public.
4. **Age confound**, *partially testable*: age is weakly decodable from the frozen features ($R^2 = 0.11$); a plausible confound worth controlling.
5. **Child-level leakage**, *addressed by design*: one scan per child and child-grouped CV.
6. **Scan count vs. child count**, *a counting distinction, not an error*: the public release is longitudinal and lists multiple visits per child, so a T1 count (the release lists on the order of 279) is larger than the number of unique children (96). We note this only to explain why our n is child-level; it is not a claim that any prior count was wrong.

5.4 What this means in practice

Method choice materially changes the reported "important regions." Three practices follow: report occlusion-based faithfulness (deletion/insertion) rather than trusting a map's appearance; run a randomized-weight sanity check and discard methods that fail it (here, Integrated Gradients); and never report a single method's regions without a corrected statistical test.

6 Limitations

$n = 96$; a single proxy task; one cortical atlas; and no PAE data. The backbone weights are the public UK Biobank brain-age release of Peng et al. (2021) (Peng, 2021); we report the checkpoint used rather than relying on an unverified source. The proxy head is weak (≈64% CV accuracy), and faithfulness and sanity verdicts are properties of the model being explained: a stronger classifier could rank the methods differently, so the rankings here should be read as holding for this modest-accuracy model, not as universal. At $n = 96$ the study is powered for medium-to-large region effects; small effects may be inconclusive (H3 and H4 do not depend on effect size). Two tests we did not run bound the interpretation. First, we measured cross-method *agreement* but not true map *stability*: we did not quantify how a single method's map varies under small input perturbations or across random seeds, so claims about reproducibility within a method are out of scope. Second, our sanity check uses complete model randomization; we did not perform the layer-by-layer cascading randomization of Adebayo et al. (2018), which would localize where in the backbone a method's dependence breaks down. Crucially, this work validates an explanation protocol on an age-group proxy in public controls. It says *nothing* about alcohol exposure, FASD, or the original paper's clinical regions, which require the private PAE data. The deliverable is the method verdict, not the accuracy.

7 Conclusion

On a frozen pretrained SFCN, nine explanation methods gave sharply different accounts of what the model uses. Occlusion-based ReX was the most faithful; Integrated Gradients failed a basic sanity check; methods agreed only weakly; and a corrected masking test isolated 15 cortical regions under zero-fill, though not under the mean-intensity fill. The practical message for neuroimaging XAI is that attribution maps must be validated for faithfulness, model-dependence, and anatomical significance, before any of them is read as evidence about the brain.

Reproducibility

Data: public Calgary Preschool controls (OSF fork `5kz2p`, CC-BY 4.0). Code, the pre-specified plan, all figures, and the full per-region p-value log are in the project repository: `https://github.com/HoussemDs/brain-xai-faithfulness-audit`. The full pipeline ran on Kaggle (GPU T4 ×2, ≈5.8 h).

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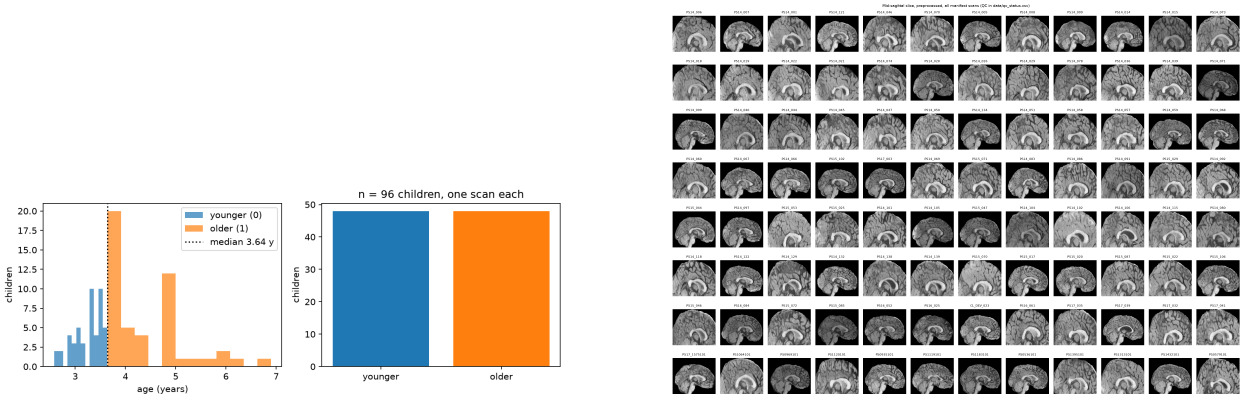
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A Supplementary preprocessing figures



(a) Age-group split at the cohort median.

(b) QC contact sheet (all 96 scans).

Figure 9: Cohort and quality control.